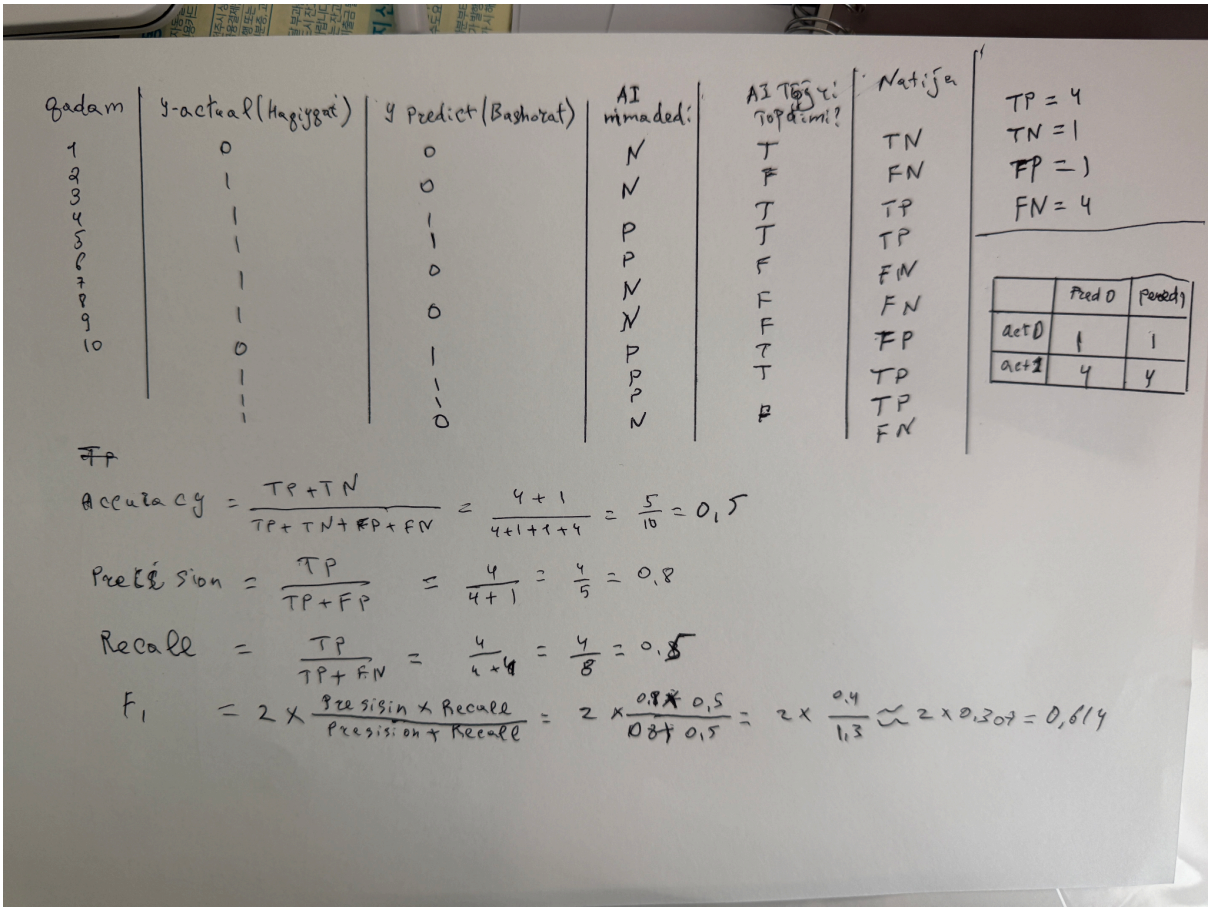
Vazifa

•Classification reportni boshqa kichik dataset orqali xar bir qismini xisoblang va koding bn tekshiring. Xisoblashlar ruchka va varoqda bolsin!

•Classificationga bilgan algorithmiz bn model yasang



Qadam	y-actual (Haqiqiyat)	y Predict (Bashorat)	AI nimaded	AI To'g'ri To'pdimi?	Natija
1	0	0	N	T	TN
2	1	0	N	F	FN
3	1	1	P	T	TP
4	1	1	P	T	TP
5	1	0	N	F	FN
6	1	0	N	F	FN
7	1	1	P	T	TP
8	1	1	P	T	TP
9	0	1	P	F	FP
10	1	0	N	F	FN

TP = 4
TN = 1
FP = 1
FN = 4

	Pred 0	Pred 1
act 0	1	1
act 1	4	4

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} = \frac{4+1}{4+1+1+4} = \frac{5}{10} = 0,5$$

$$Precision = \frac{TP}{TP+FP} = \frac{4}{4+1} = \frac{4}{5} = 0,8$$

$$Recall = \frac{TP}{TP+FN} = \frac{4}{4+4} = \frac{4}{8} = 0,5$$

$$F_1 = 2 \times \frac{Presision \times Recall}{Presision + Recall} = 2 \times \frac{0,8 \times 0,5}{0,8 + 0,5} = 2 \times \frac{0,4}{1,3} \approx 2 \times 0,307 = 0,614$$

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중단점을 추가하려면 클릭합니다. an asboblarni chaqirib olamiz
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score, f1_score

# Bizning ma'lumotlarimiz
y_actual =  [0, 1, 1, 1, 1, 1, 0, 1, 1, 1]
y_predict = [0, 0, 1, 1, 0, 0, 1, 1, 1, 0]

# Shunchaki tayyor funksiyalarga ma'lumotlarimizni beramiz
tn, fp, fn, tp = confusion_matrix(y_actual, y_predict).ravel()

print(f"TP={tp}, TN={tn}, FP={fp}, FN={fn}\n")

print(f"Accuracy: {accuracy_score(y_actual, y_predict)}")
print(f"Precision: {precision_score(y_actual, y_predict)}")
print(f"Recall: {recall_score(y_actual, y_predict)}")
print(f"F1-Score: {f1_score(y_actual, y_predict):.3f}")
[11]  ✓ 0.0s

...  TP=4, TN=1, FP=1, FN=4

     Accuracy: 0.5
     Precision: 0.8
     Recall: 0.5
     F1-Score: 0.615
```